

Multiple Choice

1. **Answer:** D

Options: A) Earth is millions of years old, and populations are unchanging.; B) Earth is millions of years old, and populations gradually change.; C) Earth is a few thousand years old, and populations rapidly change.; D) Earth is a few thousand years old, and populations are unchanging.; E) Earth is billions of years old, and populations gradually change.

Note: Correct option: D - Earth is a few thousand years old, and populations are unchanging.

2. **Answer:** D

Options: A) Epidermal cell; B) Stomatal guard cell; C) Root cap cell; D) Xylem vessel member; E) Phloem sieve tube member

Note: Correct option: D - Xylem vessel member

3. **Answer:** D

Options: A) O₂; B) CH₄ (Methane); C) CO₂; D) H₂O; E) N₂ (Nitrogen)

Note: Correct option: D - H₂O

4. **Answer:** A

Options: A) Ferns grow larger because they can perform photosynthesis more efficiently than mosses; B) Ferns grow larger due to higher nutrient content; C) Ferns grow larger due to a stronger stem; D) Ferns grow larger due to their ability to move to more fertile areas; E) Mosses are limited in size by their shorter life cycle compared to ferns

Note: Correct option: A - Ferns grow larger because they can perform photosynthesis more efficiently than mosses

5. **Answer:** A

Options: A) oxygen; B) calcium; C) carbon; D) potassium; E) phosphorus

Note: Correct option: A - oxygen

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a threshold potential'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Phloem'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Inbreeding and loss of genetic variation threaten a population.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Combustion releases energy only in the form of light, while aerobic respiration releases energy as heat.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key